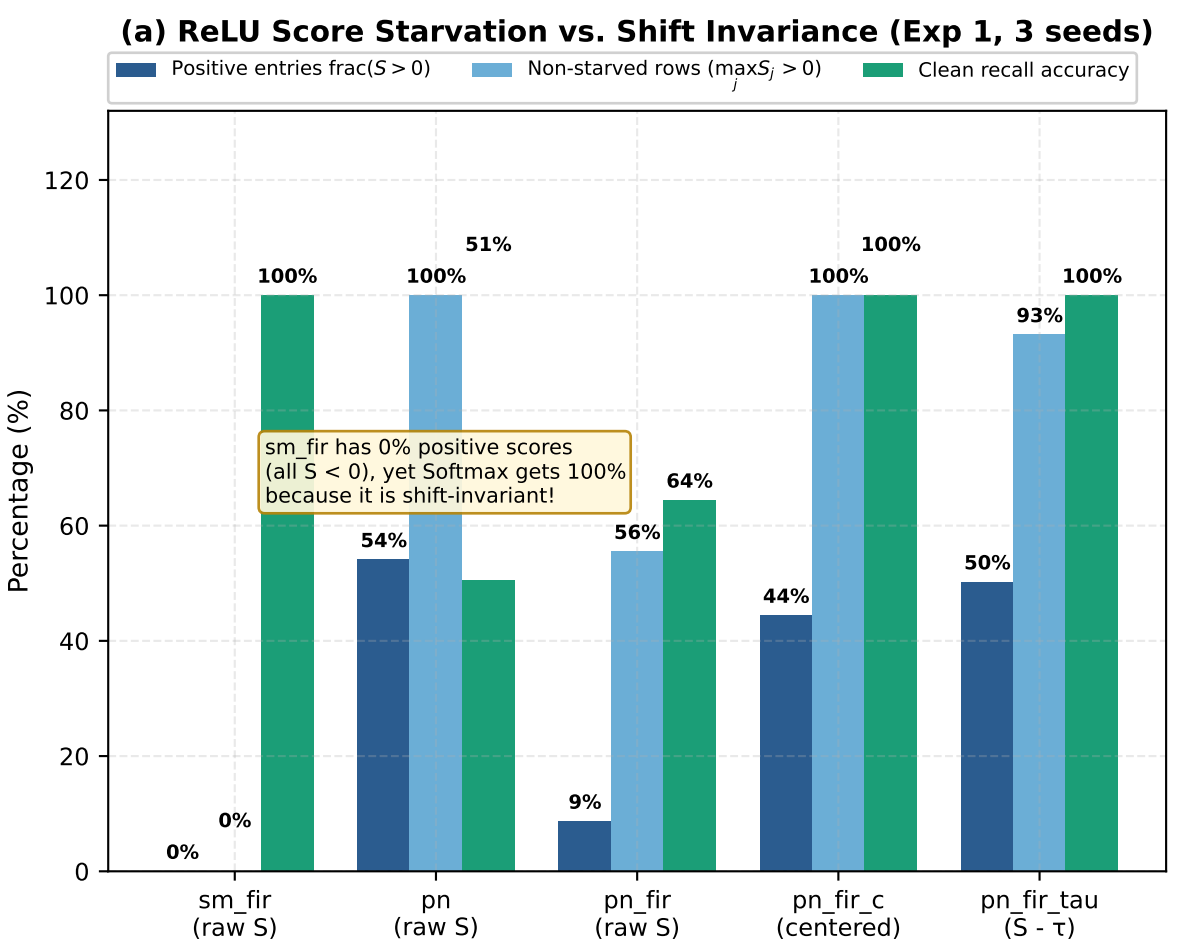
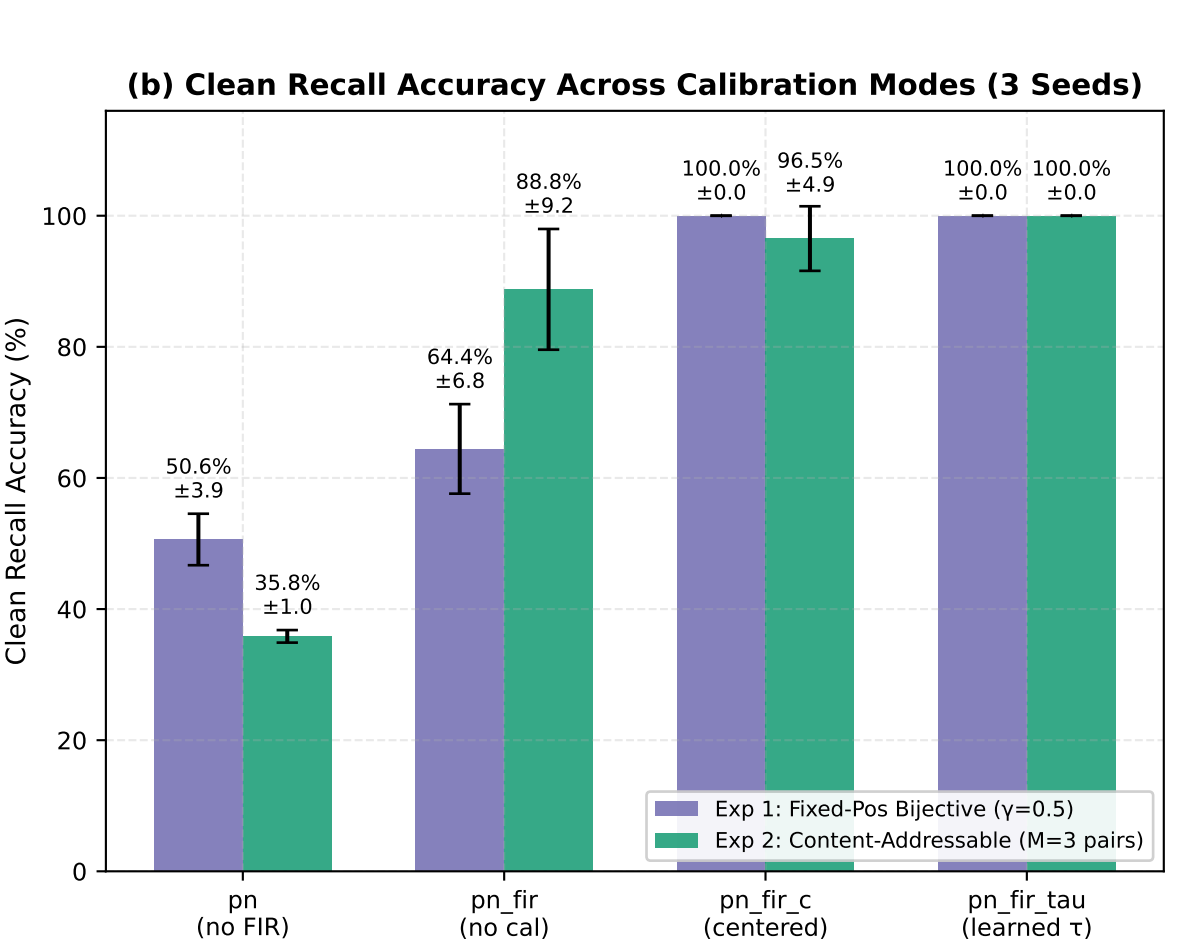

(a) ReLU Score Starvation vs. Shift Invariance (Exp 1, 3 seeds)
Positive entries frac($S > 0$)
Non-starved rows ($\max_j S_j > 0$)
Clean recall accuracy
Percentage (%)
sm_fir has 0% positive scores
(all S < 0), yet Softmax gets 100%
because it is shift-invariant!
0%
0%
100%
54%
100%
51%
9%
56%
64%
44%
100%
100%
50%
93%
100%
sm_fir
(raw S)
pn
(raw S)
pn_fir
(raw S)
pn_fir_c
(centered)
pn_fir_tau
(S - τ)


(b) Clean Recall Accuracy Across Calibration Modes (3 Seeds)
Clean Recall Accuracy (%)
50.6%
±3.9
35.8%
±1.0
64.4%
±6.8
88.8%
±9.2
100.0%
±0.0
96.5%
±4.9
100.0%
±0.0
100.0%
±0.0
Exp 1: Fixed-Pos Bijective (γ=0.5)
Exp 2: Content-Addressable (M=3 pairs)
pn
(no FIR)
pn_fir
(no cal)
pn_fir_c
(centered)
pn_fir_tau
(learned τ)